# Naga Venkata Sai Chennu

*Business Systems Analyst – AI Automation*

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## EDUCATION

**George Mason University**, Fairfax, VA — *Master of Science in Computer Science* | May 2026

## EXPERIENCE

**Graduate Research Assistant – Business Systems & AI Automation**, *George Mason University – Costello College of Business*, Fairfax, VA — Aug 2025 – May 2026

- Cut manual document processing time by ~80% by analyzing a high-volume legal document workflow, defining requirements for an AI-powered extraction pipeline (Claude API), and implementing structured JSON output with automated quality validation against human-labeled ground truth across 500+ trademark filings.
- Collected requirements from faculty and doctoral researchers, translated them into technical specifications, and built an end-to-end data pipeline—ingestion, cleaning, annotation, validation—that eliminated repetitive manual work and boosted data accuracy.
- Defined and monitored extraction-pipeline performance metrics, flagging low-confidence outputs for human review to ensure business users retained full control over final decisions.
- Architected llm-forge, a YAML-driven 12-stage automation pipeline that enables non-technical researchers to launch LLM fine-tuning jobs from a single config file, cutting time-to-experiment by over 50% through automated GPU resource orchestration (4×H100 via SLURM).
- Implemented strict Pydantic v2 schema validation at every stage with ~900 automated quality checks, catching configuration errors before runtime to prevent costly job failures.
- Served as primary liaison between business faculty and technical teams, leading weekly progress reviews; mentored and onboarded incoming graduate assistants using documented SOPs, halving their time-to-productivity.

**Undergraduate Researcher – Data-Driven Business Solutions**, *KL University – Department of CSE* — Jan 2022 – May 2024

- Designed a quantitative study surveying 468 IT professionals; performed SEM/CFA analysis that identified significant drivers (neural networks $P=0.027$, intelligent agents $P=0.000$, $R^2=0.635$) and delivered actionable recommendations to strengthen cyber-defense strategy.
- Built an end-to-end ML pipeline that combined four classifiers with ensemble methods (bagging, boosting, stacking), achieving 93% accuracy and 92% F1 validated via t-test and ANOVA—a high-performance, data-driven solution.
- Evaluated AI models (ANN vs. CatBoost) for a stress-detection use case and delivered a metrics-based recommendation that drove adoption of CatBoost (89% accuracy, 91% precision) in a web-based analytics dashboard.

## PROJECTS

**AI Business Request Intake & Jira Automation System** — *Turns vague stakeholder requests into structured requirements, user stories, and Jira-ready tickets* — 2026

Python, Streamlit, FastAPI, Claude API, PostgreSQL, Jira API (mock), LangChain

- Mapped the as-is request-intake process and catalogued 12 common request patterns to define target artifacts—BRD, FRD, user stories, acceptance criteria, risk notes, and priority scores.
- Built an AI intake assistant that transforms a free-text request into all those structured outputs in under 60 seconds, paired with a mock Jira API that auto-creates implementation tickets.
- On sample requests, cut requirement-clarification turnaround by ~70% and produced reusable artifacts—a BRD template, user-story backlog, UAT test cases, and a process map.

**AI Automation ROI & Process-Optimization Platform** — *Scores manual processes by effort, risk, feasibility, and ROI to guide automation strategy* — 2026

Python, Streamlit, Plotly, Claude API, PostgreSQL

- Designed a 6-dimension scoring framework covering time, cost, risk, feasibility, AI-suitability, and integration complexity, and inventoried 25+ representative manual workflows.
- Built an interactive platform where users enter a workflow and receive automated scoring, ROI estimates, AI-generated automation proposals, risk assessments, and roadmaps.
- Modeled $120K+ in potential annualized savings on sample data and packaged the results as a leadership-ready prioritization model, risk matrix, and stakeholder deck.

**Healthcare Workflow Automation & EMR Data-Extraction System** — *Compliance-aware extraction, validation, and routing of healthcare operations data* — 2026

Python, FastAPI, PostgreSQL, Claude API, Streamlit

- Analyzed EMR-style extraction workflows, mapped 5 exception types, and designed a HIPAA-aware architecture that runs entirely on synthetic patient data.

## PUBLICATIONS

5 peer-reviewed papers (2 first-author) plus 1 patent, 50+ citations across AI/ML, data analytics, cybersecurity, and NLP.

## CERTIFICATIONS

NVIDIA Certified Associate – Generative AI and LLMs (Feb 2026) • AWS Certified AI Practitioner (Nov 2025) • Anthropic – Claude Code in Action (Dec 2025)

## SKILLS

**Business Analysis & Process Automation:** Requirements Gathering & Documentation, Process Mapping (As-Is/To-Be), Workflow Automation Design, Vendor / Technology Evaluation, Success Metrics & Dashboards

**AI & Data:** LLM APIs (Claude, GPT), Prompt Engineering, Statistical Analysis (CFA, SEM), Ensemble ML & Model Validation

**Tools & Platforms:** Python, SQL, Jira, SLURM, Pydantic v2